An Efficient Image Forgery Detection Framework using Transfer Learning Models

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Abstract— In recent years, digital images have become common in modern life in various applications. This leads to be attractive to attackers to modify them for malicious activities. The process of modification of digital images is known as image forgery. There is a wide range of software and hardware tools available for modifying and producing them. It is of utmost importance to understand how authentic and forged images can be detected and classified since digital image forgery can be extremely damaging. Since the advent of digital photography, there have been new and creative ways to forge images. When images are altered by forgeries, it affects how the viewer perceives the scenario that is being shown, which can have negative effects if done maliciously. Therefore, there is a serious need to provide a method to detect the modification of digital images. Thus, this paper provides an efficient image forgery framework using transfer learning models for the effective detection of image forgeries. From the results, the proposed method proved to be most accurate in detecting image forgeries on a dataset containing a mixture of 220 authentic and forgeries images, these models as RaseNet50, VGG16, Xception, Densenet201, MobileNet, and NasNetMobile. The RaseNet50 gives the highest accuracy of 96%.to detect image forgeries in a real-time manner.

Keywords— Classification, Deep Learning, Copy-Move Forgery, Tampered Images, Transfer Learning, RaseNet50, VGG16, Xception, MobileNet, and NasNetMobile.

1. Introduction

In recent years, the use of videos and photos has increased because they are the primary sources of information. Images are frequently used to communicate information to various recipients. They are used in a variety of situations, including military operations, intelligence, courtroom evidence, social media, newspapers, misinformation in the media, defamation of well-known figures, and many other applications. In addition to the unsecured methods for image circulation, image manipulation may compromise the authenticity of an image. Authentication mechanisms have been created to secure picture communication. Active and passive authentication falls into two kinds[1]. Active authentication is done by putting a watermark on the original image material before using it, digital image cryptography, or a digital signature.

The ability to match the test image to the original image's content is required for active authentication [2]. Data-embedding algorithms and various payload partitioning techniques are both related to active authentication, which is used in the security of colored image steganography [3, 4]. A technique for separable data hiding in encrypted images has been developed for active authentication using discrete Fourier transform and compressive sensing. [5]. This method ensures recovery and offers a configurable payload. The image's original content is hidden during passive authentication. The test image is examined without any prior knowledge of the original content [6]. Digital image forgery detection is a significant problem. Finding proof of a fake is done by looking for unique characteristics, features, or disorders. The original image's statistical, geometrical, or physical qualities must be homogeneous for the forgery detection method to work, but image manipulations lose this homogeneity [7]. Digital image forgeries come in several forms, such as image splicing, morphing, and copy-move forgery [8]. Image splicing is the process of combining two or more images to make a new image containing features from the source images [9]. The technique of blending separate shapes from different pictures to create a new shape is typically referred to as image morphing [10]. The process of image retouching involves altering a scene or photograph to highlight or obscure certain characteristics, for example by adjusting the lighting, contrast, color, or background [11]. Resampling an image involves altering one or more of its dimensions to convey a different meaning to the audience [12]. To replicate or conceal one or more objects and provide a fake view, copy-move forgery involves copying and pasting a region or area from one location in an image onto another location in the same picture [13]. Examples of several types of digital image forgeries are shown in Fig. 1.

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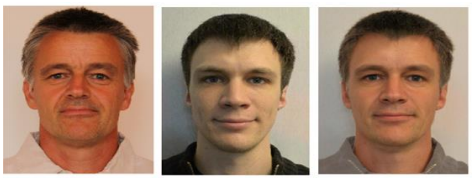
Because it is easy to use and hard to detect, copy-move forgery is the most prevalent sort of digital image counterfeiting. Its difficulty in being detected varies because of its slight effect on the image.

The main objective of image forgery detection is to classify the input image into original and fabricated images.

The suggested structure is divided into two parts to achieve this: the first part performs pre-processing, and the second part is the training models, which extract features from the pre-processed images and perform dual classification of images as real or fake.

In the suggested system, we performed image pre-processing since training a convolutional neural network on unprocessed images would probably lead to poor classification performance and low accuracy. A faster training process also requires image pre-processing. Thus, input images are initially downsized to a defined size during the image pre-processing step without any cropping because large-sized images increase computational complexity, and the system takes longer to converge. Finally, to detect image forgeries, we compare transfer learning models such as RaseNet50, VGG16, Xception, Densenet201, MobileNet, and NasNetMobile. In a blind experiment comprising 220 real and fake images, these techniques identified image forgeries with an observed accuracy of 96%.



(a)image morphing: the left and middle images are originals, while the right is the tampered image.

(b)copy-move forgery: beginning from the left, the first image is the original image followed by its tampered image.

Fig 1. Examples of Digital Image Forgery Types

The structure of this paper is organized as follows. Section 2 displays the relevant work in this important field, while Section 3 examines the suggested method for image forgeries. The experimental results achieved with the suggested system on the MICC 220 dataset with various attack types are shown in Section 4. Section 5 presents the work's concluding remarks.

2. Previous Studies

Digital image manipulation research has grown significantly during the last few years. To develop forensic tools for digital images. This section will discuss recent developments in the fields of image forensics and forgery detection.

In [14], They outlined a convolution neural network based on deep learning that performs copy-move.

In [15], they proposed a Convolution Neural Network-based deep learning to identify image tampering. In [16], they proposed a deep learning method was developed to identify automatically generated fake images of faces produced by generative adversarial networks (GAN).

In [17], they suggest a model that will be created making use of a better discrete cosine transform.

In this study, the decomposition of quantization coefficients is based on the Discrete Wavelet Transform (DWT). The model's goal is to find copy-move picture forgeries. In terms of scaling, rotation, and translation, it is inefficient. It only works for straightforward picture forgeries. In the future, we'll be trying to make sure that extracted features are resistant to forged caused by rotation, translation, and scaling. The author of this paper [18] proposed a newly constructed digital image forensic method. This study is concerned with geometric inconsistencies.

To produce a modified image, fake reflections are added to the image. Some photo-editing techniques, such as color manipulation, resampling, and lossy compression, do not affect the proposed method. In [19] The authors outline a copy move manipulation detection model. We employ the MICC-F220, MICC-F8multi, and CoMoFoD datasets in this experimental study. the performance of existing key-point-based and block-based copy move manipulation identification tools has been improved. This technique is using transformations, including rotation, scaling, and translation. In [20], The author examined many copy-move manipulation identification techniques and identified the ones that worked best with various photo properties like rotation, scale, translation, and JPEG compression. additionally, lists each method's advantages and disadvantages. The author compares block-based and key point-based techniques to identify copy-move manipulation and concludes that key point-based approaches outperform block-based techniques in terms of performance and computation time.

In [21], The author presented an experimental study for CNN-based image tampering identification. This neural network also detects splicing forgery. A copy-move manipulation identification approach is described in [22].

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The detection of copy-move photo tampering is performed using the feature-based SIFT approach. The experimental work resolves the issue of unimportant key points in key point selection. In [23], The authors suggested a simple yet efficient technique for copy-move detection.

Using the MICC-F220 dataset, they presented a pre-trained AlexNet model in [24] that was used to extract 4096-element deep feature vectors from the input images

(note that the feature vector is obtained by removing the classification layer of the AlexNet network).

After the collected features are loaded into an SVM model, the entire image is binary classified as authentic or fraudulent. There are two phases to the training process. Initially, the pre-trained AlexNet CNN is used to extract features from both the real and the fake photos. A copy-move forgery detection method is provided in [25]. There is no pre-computed separation of features; it is entirely DL-based.

This method simply produces a global classification of the image, not the localization of the forged regions.

Four benchmark datasets have been used to assess it: SATs-130, MICC-F220, MICC-F600, and MICC-F2000. In [26], they proposed a new image forgery detection system using CNN to automatically learn hierarchical representations from the input RGB color images. To determine if an image is authentic or forged, the method discussed in [27] also uses a CNN. However, this method used a shallow CNN model, consisting of only two convolutional layers, as opposed to the previously stated methods, which used deep learning networks with several layers. Additionally, dimensionality reduction was accomplished without the need for max-pooling steps by taking advantage of big convolutional filters, the sizes of which were 32 by 32 for the first layer and 20 by 12 for the second.

3. Proposed Forgery Detection System

The main objective of image forgery detection is to classify the input image into original and fabricated images. Detailed stages of the proposed system are shown in Fig 2 as follows:

Stage 1: A total of 220 images were randomly extracted from the MICC 220 dataset. These images are used for the training and testing stages of the model.

Stage 2: Image pre-processing is essential since training a convolutional neural network on raw images is likely to result in poor classification performance and low accuracy.

Furthermore, image pre-processing is essential for accelerating training. Large-sized images increase computational complexity and cause the system to take longer to converge, so during the image pre-processing stage, input images are first reduced to a set size without any cropping.

Stage 3: In particular, 70% of images were used for training and the remaining 30% for testing.

Stage 4: Transfer Learning Models such as RaseNet50, VGG16, Xception, Densenet201, MobileNet, and NasNetMobile are applied for image forgery detection.

Stage 5: To perform and evaluate the performance, we used the following metrics: accuracy, sensitivity, and specificity.

These metrics are used to assess the result of the provided framework is predict whether the image is original or modified in a real-time manner.

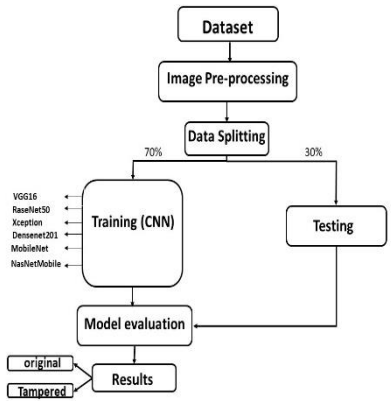

Fig 2. Proposed Image Forgery Detection Framework.

4. Experimental Study

This section explores and presents the experimental environment, dataset, and assessment metrics. Finally, explore the obtained results of the proposed system.

a. Experiment environment

The experiments were conducted on a Windows 10 PC with an Intel Core i5 4200U 64-bit CPU running at 1.60 GHz and 2.30 GHz and 4 GB of RAM. Python 3.8 is used to implement the transfer learning models, and Keras and TensorFlow backend toolkits are used.

b. Dataset

The MICC-F220 dataset has manipulated sections that account for 1.2% of the entire image [28]. The forgeries consist of both plain copy-moves and copy-moves that have been rotated, scaled, or both. All of the copied parts are rectangular. The description of this dataset is shown in Table 1.

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Table 1. Dataset Description.

Dataset	composition	Size of images	Size of tampered region
MICC-F220	It consists of 220 images divided into 110 tampered and 110 original	Between 722 x 480 and 800 x 600 pixels	The region represents 1.2% of the whole image

c. Evaluation Metrics

The performance evaluation of transfer learning models in a precise image forgery classification and prediction task is measured by several statistical and mathematical models.

These performance evaluation metrics such as accuracy, specificity, and sensitivity, are calculated as per in Equations (1)-(3) respectively [29-40]:

$$\text{Accuray} = \frac{TP+TN}{TP+FP+FN+TN} \quad (1)$$

$$\text{Specificity} = \frac{TN}{TN+FP} \quad (2)$$

$$\text{Senstivity} = \frac{TP}{TP+FN} \quad (3)$$

d. Results Analysis

The results of the transfer learning model such as RaseNet50, VGG16, Xception, DenseNet201, MobileNet, and NasNetMobile are applied to the MICC-F220 dataset. The number of training sessions or epochs at which the best performance is achieved has been tested in several trials employing various numbers of epochs.

At epoch 15, the RaseNet50 model achieved the highest accuracy of 96% as explained in Table 2.

The results of the tests carried out on the MICC-F220 Dataset for various epoch counts are displayed in Tables 2, 3, and 4. Results are explored as well in Figs. 3, 4, and 5. These results demonstrate that as the number of epochs increases, the accuracy, sensitivity, and specificity values increase as well, reaching 96% at 15 epochs.

Table 2. Accuracy result for image forgery using transfer learning models

Epochs	Accuracy%	RaseNet50	VGG16	Xception	DenseNet201	MobileNet	NasNetMobile
5		89	73	82	80	93	89
10		91	67	82	96	84	91
15		96	74	89	93	91	91
20		93	85	89	93	93	89

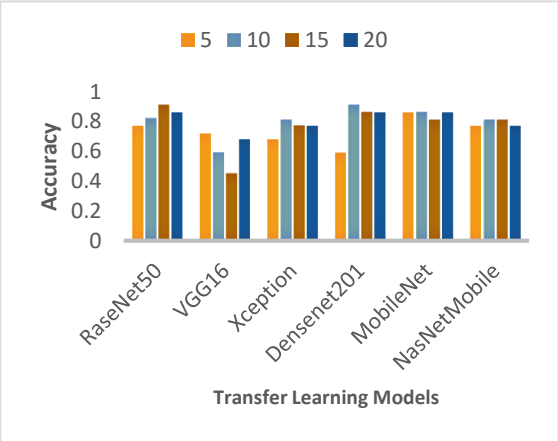

Fig 3. Accuracy result for image forgery using transfer learning models.

Table 3. Sensitivity results for image forgery using transfer learning models

Epochs	Sensitivity%	RaseNet50	VGG16	Xception	Densenet201	MobileNet	NasNetMobile
5		100	75	95	100	100	100
10		100	75	83	100	84	100
15		100	100	100	100	100	100
20		100	100	100	100	100	100

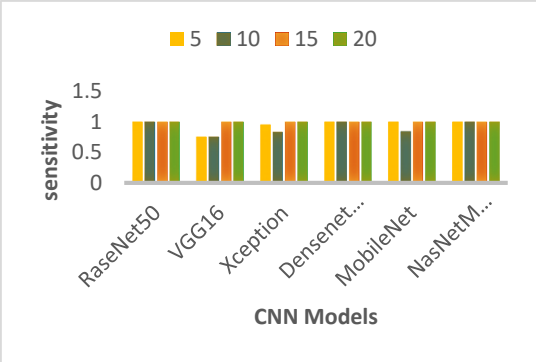

Fig 4. Sensitivity result for image forgery using transfer learning models.

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Table 4. Specificity result for image forgery using transfer learning models

Epochs	Specificity%	RaseNet50	VGG16	Xception	Densenet201	MobileNet	NasNetMobile
5		77	72	68	59	86	77
10		82	59	81	91	86	81
15		91	45	77	86	81	81
20		86	68	77	86	86	77

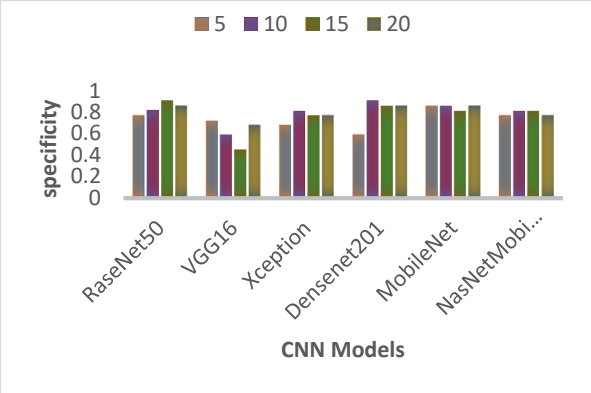

Fig 5. Specificity result for image forgery using transfer learning models.

e. Complexity analysis

Table 5. shows the execution time result among transfer learning models

Methods	Testing time(sec)
ResNet50V2	22.00
VGG16	68.00
Xception	34.00
Densenet201	30.00
MobileNet	8.00
NasNetMobile	14.00

From the obtained results given in this table, it seems that the MobileNet model is the best model for consuming processing time. This indicates that this model is less complex as compared to other models.

5. Conclusion

Meanwhile the emergence of digital photography, novel and inventive methods for fabricating images have arisen. The manipulation of images through forgery can significantly alter an observer's perception of the depicted scene, potentially leading to negative consequences if done with malicious intent. Therefore, there exists a pressing need for a reliable method to identify modifications in digital images.

Accordingly, this paper presents an effective image forgery detection framework leveraging transfer learning models.

The results demonstrate that the proposed approach is highly accurate in detecting image forgeries within a dataset comprising 220 images, encompassing both authentic and forged ones. The models utilized include ResNet50, VGG16, Xception, DenseNet201, MobileNet, and NasNetMobile. A comparative analysis is made using various pre-trained models– VGG16, ResNet50, MobileNet, DenseNet201, NasNetMobile, and Xception on the MICC-220 dataset. Based on the analysis of the results obtained, it can be concluded that ResNet50 performed best for the MICC-220 dataset with 96% accuracy at epochs 15, and denseNet201 produced 96% accuracy for epochs 10. VGG16 saw its best accuracy of 85% on the epochs 20.

Xception performed best at epochs 20 with 85% accuracy and Xception produced 89% accuracy. MobileNet produced 93% accuracy for the epochs 20 and NasNetMobile produced 91% accuracy for the epochs 15. The results reveal that in a completely blind experiment using a mixture of 220 original and forged images, the best performance was obtained with 15 epochs for the MICC-F220 dataset with 96% accuracy.

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